

Lithium based Ionogel as Solid State Electrolyte: Dynamics of Confined Ionic Liquid, a Neutron Diffusion Study

C. V. Cerclier^a, J.M. Zanotti^b, J.P. Embs^c, J. Le Bideau^{*a}

^a Institut des Matériaux Jean Rouxel (IMN), Université de Nantes CNRS, Nantes 44322, France

^b Laboratoire Léon Brillouin, CEA-CNRS, Gif sur Yvette 91191, France

^c Laboratory for Neutron Scattering and Imaging, PSI, Villigen 5232, Switzerland

Obtaining solid-state electrolytes with good electrochemical performances remain challenging. Ionogels, i.e. solid host network confining an ionic liquid, are promising as they keep the macroscopic properties of the confined liquid. However, confinement of an ionic liquid can imply important changes in its molecular dynamics, depending on the route of synthesis and on the confining network. We studied this effect on an imidazolium based ionic liquid with its Lithium salt confined in a biopolymer-silica matrix. Dynamics of bulk and confined solution was probed by quasi-elastic neutron scattering (QENS). The scattering function was modeled successfully with two Lorentzian functions in both cases and revealed a weakly slowed dynamics of imidazolium-based ionic liquid inside the polymer-silica host network.

Introduction

Ionic liquids (IL) have noteworthy and well-known properties such as very low volatility, good chemical and thermal stability, wide electrochemical window and good ionic conductivity. These properties make them particularly promising for developing safer electrochemical energy storage devices (solar cells, double layer capacitors and lithium batteries) (1). One way to take advantage of these properties is to confine ionic liquids in solid-state materials, like polymers, mesoporous oxides or composite matrices. Such material can be obtained by swallowing of IL into a previously prepared host network, or by a one-step sol-gel process leading to ionogels (fully mineral or hybrid organic-inorganic)(2,3). The resulting silica-based material, with or without added polymer, is a solid with liquid-like properties (2) whose efficiency as solid electrolyte has been demonstrated in lithium batteries (4).

Previous studies show that phase transitions of silica-confined IL are modified, resulting in liquid-like behavior below the solidification temperature of neat IL (2,5,6). Besides, confinement is a means known to modify dynamics of molecular liquid. Literature reports either fastened (7) or lowered (8) molecular dynamics inside different kinds of host network.

In this work, we have studied dynamics of 1-butyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide containing 0.5 M LiTFSI (BMImLiTFSI) by means of quasi-elastic neutron scattering (QENS). The results have been compared to its counterpart confined inside a composite biopolymer-silica host network. This network

comprises siloxane hydroxypropyl methyl cellulose (Si-HPMC) covalently linked to silica nanofibers (NFs) via siloxane bonds. These silica NFs are around 50 nm in diameter and 500 nm long.

Experimental details

Ionic liquid and ternary synthesis

1-Butyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide BMImTFSI was synthesized as a colorless transparent liquid according to the literature (9). 1-methylimidazole was reacted with a slight excess of 1-bromobutane in a roundbottom flask and left refluxing for 2 h to produce 1-butyl-3-methylimidazolium bromide. The corresponding TFSI salt was then prepared by metathesis of [BMIm][Br] in water with Lithium bis(trifluoromethylsulfonyl)imide (LiTFSI). The obtained ionic liquid was checked by ^1H -NMR. The ternary solution BMImTFSI with lithium salt at 0.5 M was obtained by adding LiTFSI (0.7 g) to the IL (7 g), namely BMImLiTFSI.

Ionogel Synthesis

The hybrid ionogel was synthesized by a sol-gel process involving the mixing of 2 solutions, as described previously (5). The solution A was obtained by dissolving Si-HPMC polymer in 0.2 M NaOH aqueous solution (30.9 mg ml⁻¹, pH 12.5); then two dialyses with molecular weight cut off at 6–8 kDa were performed in 0.09 M NaOH_{aq}. The dialyses eliminate the non-grafted 3-glycidoxypolytrimethoxysilane, used for siloxane grafting onto the HPMC. In solution B, 60 mg silica NFs were dispersed in 2.36 mL of a mixture of acetonitrile and formic acid to which 1 g of BMImLiTFSI was added. The ionogel sol precursor was then obtained by mixing 1.4 g of solution A with solution B. Gelation of the sol occurred in few hours. This route allowed to reach an IL-Li content as high as 91 wt%, within a confining network made of 57/43 wt% of Si-HPMC and NFs resp : the composition [polymer / silica] / IL-Li is thus summarized [57 / 43] / 91.

QENS experiments

QENS experiments have been performed on the cold neutron time-of-flight spectrometer FOCUS (10) at the Swiss spallation neutron source SINQ, Paul Scherrer Institute, Villigen, Switzerland. For these experiments, we used neutrons with an incident wavelength of $\lambda = 5.76 \text{ \AA}$. The energy resolution at this wavelength is $\Delta E \approx 45 \text{ \mu eV}$. Spectra were recorded for 4 different temperatures $T = 30, 233, 300$ and 333 K . Data obtained for samples at 30 K were used as a resolution function. We used flat sample containers made of aluminum which suits with 200 μm thick samples. This thickness guarantees neutron transmission of >90% and thus suppresses multiple scattering. The measured intensities of a vanadium standard with the same shape as the sample have been used to calibrate the detectors. We employed measurement times of about 6 h for each temperature. The DAVE software package (11) was used for data reduction and analysis. Data were binned into 14 groups of constant Q values; the energy range was set to $\pm 1 \text{ meV}$.

Results and Discussion

Quasi-elastic neutron scattering (QENS) is an appropriate technique to study local dynamical processes in condensed matter. It allows investigating diffusion, relaxation, localized processes (such as reorientations) and jump diffusive motions; one can extract quantitative parameters such as diffusion coefficients, jump lengths and relaxation and reorientation times. QENS is very sensitive to samples containing hydrogen due to the large incoherent neutron scattering cross section of proton. The IL studied here, BMImLiTFSI, contains protons only on the cation, so the whole study presented here gives information on the dynamics of the imidazolium derivative, without any direct indication of the dynamics of the anion.

QENS experiments on bulk and ionogel-confined BMImLiTFSI were realized at the following temperatures: 233, 300 and 333 K. In this temperature range, the ternary IL BMImLiTFSI is in a liquid phase. The Figure 1-A presents the $S(Q, \omega)$ spectra of the bulk solution at $Q = 1.29 \text{ \AA}^{-1}$. The signal broadened upon increasing temperature, which corresponds to the quasi-elastic part of the spectra which becomes more and more important from 30 K (pure elastic signal) to 333 K. The analysis of elastic intensity has allowed determining the mean-square displacement at these four temperatures (Figure 1-B). A break in slope seems to occur below 233 K, highlighting the transition between pure vibrational motions and translational diffusion. The observed behavior is coherent with calorimetric measurements (DSC) made on BMImLiTFSI which only shows a glass transition at 191 K (data not shown).

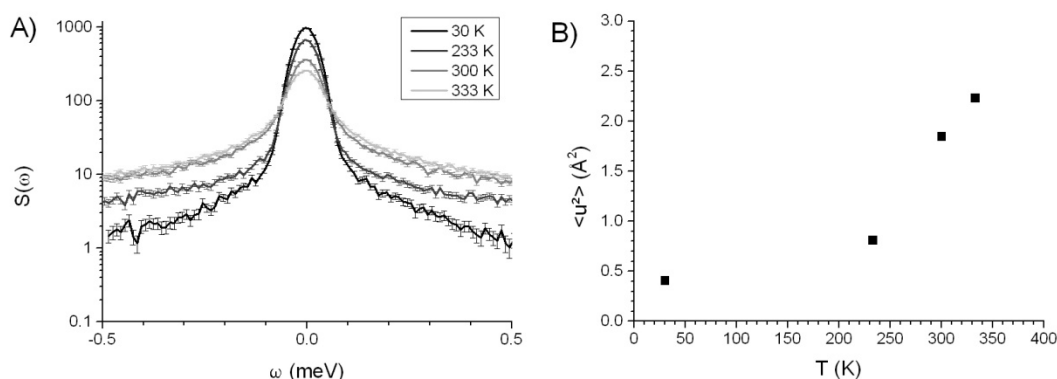


Figure 1. A) QENS spectra of BMImLiTFSI obtained at $T = 30, 233, 300$ and 333 K for $Q = 1.29 \text{ \AA}^{-1}$. B) Mean-square displacement of BMImLiTFSI as a function of temperature.

At 300 and 333 K, the scattering function of bulk BMImLiTFSI has been fitted satisfactorily with the combination of two Lorentzian functions. This model is usually used for liquid phases for which the narrowest Lorentzian function, via its full width at half maximum (FWHM), characterizes the translational diffusion of the cation (Figure 2). However this model does not match for the scattering function measured at 233 K. Dynamics of BMImLiTFSI should be too low at this temperature with regard to the energy resolution at $\lambda = 5.76 \text{ \AA}$ ($45 \text{ \mu eV}$).

In the case of simple translational diffusion, the FWHM of the translational term has a quadratic dependence on Q . But at large Q , dynamics is often described by the rapid jump diffusive model (equation 1).

$$FWHM = \frac{2\hbar D Q^2}{1 + D Q^2 \tau_0} \quad [1]$$

Where D is the diffusion coefficient and τ_0 is the characteristic residence time between jumps. This last model allowed fitting data with an acceptable precision. Values of fitted parameters are reported in Table 1. Value of diffusion coefficient of BMImLiTFSI is of the same magnitude than that of ILs in their liquid phase (12-14). In our case, the presence of lithium salt should slightly lower the dynamics of BMIm cation of pristine IL. This phenomenon is consistent with others experimental facts such as the increase of both density and viscosity with small amount of Lithium ions.

In the case of IL confined in the polymer-silica host network, two Lorentzian functions also fitted well to the experimental data recorded at 300 and 333 K. The Figure 2 presents the FWHM extracted from these fits as a function of $1/Q^2$. For the same temperature, the FWHM of confined IL is always smaller than these of bulk one. It is an indication of a slowing down of the dynamics of BMIm cation. The values of diffusion coefficient D and of characteristic residence time τ_0 are consistent with such phenomenon (Table 1). It is worth to point out that, within the polymer-silica host network, the slowing down of the imidazolium cation remains very small and sometimes within the experimental error. However, dynamics of ILs appears to be more complex than simple molecular liquids usually described by a two-Lorentzian functions model. In particular, conformational motions of alkyl functional group, like rotational reorientation or dihedral relaxation, can interfere with the signal stemming from the global cation's dynamics (13). This is especially true when the length of the alkyl side-chain increases. This makes the precise determination of diffusion coefficients more difficult. Nevertheless, one can observe for the characteristic residence time an increasing trend upon confinement, which is quite expected, based on interaction at the interface (15).

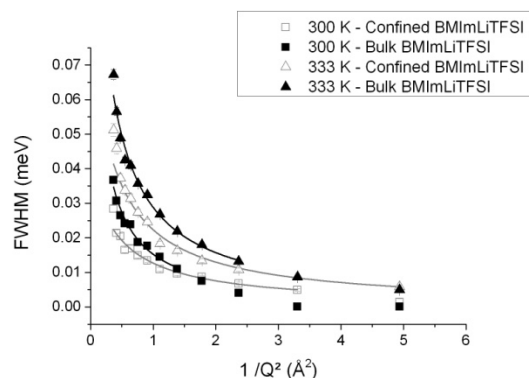


Figure 2. Full width at half maximum (FWHM) versus $1/Q^2$ at 300 K (squares) and 333 K (triangles).

TABLE I. Values of the diffusion coefficient D and the characteristic time τ_0 of bulk and confined BMIm in BMImLiTFSI at 300 K and 333 K.

	Temperature (K)	D ($10^{10} \text{ m}^2 \cdot \text{s}^{-1}$)	τ_0 (ps)
Bulk BMImLiTFSI	300	$1.30 (\pm 0.07)$	$10 (\pm 3)$
	333	$2.62 (\pm 0.06)$	$7 (\pm 1)$
Confined BMImLiTFSI	300	$1.41 (\pm 0.05)$	$33 (\pm 3)$
	333	$2.33 (\pm 0.06)$	$16 (\pm 2)$

Conclusion

This work has shown that, in the ternary IL BMIm TFSI with lithium salt, BMIm cation presents diffusion coefficients and characteristic times comparable to other ionic liquids already reported in literature. Its confinement within a biosourced polymer-silica matrix host network allowed reaching a very high loading level with a one-step preparation. It has been observed that the molecular dynamics of BMIm cation slowed down only slightly. These results highlight the advantage of biosourced solid ionogel exhibiting liquid-like properties.

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